

Do individuals share a causal dynamical operator?

Zero-shot transfer of intervention responses between animals

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Abstract

Latent neural dynamics are strikingly similar across individuals performing the same behaviour, and a growing family of methods aligns recordings across animals into a shared representation. Those results are *descriptive*: they show that observed activity occupies a common geometry. They do not establish that different individuals share a common *causal* operator – that the same physical intervention would move each brain the same way. We formalise that stronger claim as a prediction problem: after observing only *normal, unperturbed* activity from a new animal, predict the full time-resolved neural and behavioural response to an intervention that animal has never received. We introduce CADENCE, a hierarchical controlled latent dynamical system that separates a species-invariant dynamical operator, a species-invariant causal (intervention) operator, animal-specific observation maps and animal-specific residual dynamics; we show that the new animal’s observation map – and hence its predicted intervention response – is identified from unperturbed activity alone up to the symmetry group of the shared flow. In teacher-RNN simulations with known ground truth, zero-shot transfer is positive when animals genuinely share dynamics ($\Delta R^2 = +0.230$ and $+0.182$ in two such regimes, against ceilings of 0.983 and 0.929) and negative between independently trained networks that share only the task (-0.104), with every negative control below zero; with five simulated animals per regime these estimates are imprecise, so we treat them as existence checks rather than measurements. We then apply the framework to 6 mice receiving parameterised intracortical microstimulation of somatosensory cortex during a detection task (48 sessions, 949 single units, 16,532 stimulation trials; DANDI:001868), holding out whole animals and, in the strictest condition, also deleting the stimulation amplitude from every training animal. The answer is granularity-dependent and, we argue, biologically informative. The time-resolved behavioural response transfers strongly to a completely new animal ($\Delta R^2 = +0.577$, 95% CI $[+0.471, +0.669]$, 0.60 of the noise ceiling, $p = 7.1 \times 10^{-15}$, the floor of the exact paired test at $n = 48$), and remains positive when the amplitude is also unseen ($\Delta R^2 = +0.321$). Coarse-grained neural responses transfer once the animal’s responsiveness is itself predicted from its spontaneous activity, as linear-response theory suggests it should be: the population response reaches $\Delta R^2 = +0.289$ $[+0.174, +0.397]$ ($p = 1.0 \times 10^{-5}$), about half the $+0.571$ attainable with the gain known exactly. Single-unit responses do not transfer ($\Delta R^2 = +0.003$, $p = 0.916$) even though they are well estimated (ceiling 0.914) and largely predictable within a session ($\Delta R^2 = +0.329$). We then localise that failure precisely: holding the shared operator completely fixed and granting the held-out animal only *one scalar per unit* raises ΔR^2 from $+0.003$ to $+0.405$ ($p = 7.1 \times 10^{-15}$), and a per-unit scalar cannot invent structure in time or across conditions. The shared causal operator is therefore already substantially correct; what is missing is the animal-specific map between individual neurons and the conserved state. Responsiveness turns out to be legible in spontaneous activity at the level of the animal ($r = +0.415$) but not of the neuron ($r = +0.174$), and for microstimulation we show why: the stimulus recruits a spatially diffuse, implant-specific set of cells ($r(|\Delta z|, \Delta \text{rate}) = -0.013$). Shared neural dynamics is causal, but only above the single-neuron scale.

1 Introduction

Populations of neurons recorded in different individuals performing the same task occupy remarkably similar low-dimensional manifolds, and the flow on those manifolds looks similar too. Methods that exploit this have become standard: local flow fields can be embedded in a common space [1], recordings can be stitched across sessions and subjects to yield latent dynamics that generalise behavioural decoding [2–4], and alignment techniques recover correspondences between populations that share no neurons [5–7].

Every one of these results is a statement about *observed* activity. A shared manifold, and even a shared vector field fitted to spontaneous or task-driven activity, is compatible with individuals whose *causal* structure differs: two

dynamical systems can produce the same trajectories under the conditions sampled and yet respond differently to the same push. The distinction matters practically – it decides whether a stimulation protocol calibrated in one brain can be expected to act the same way in the next – and conceptually, because “shared dynamics” is usually intended as a claim about mechanism.

We therefore ask a question that a purely observational alignment cannot answer:

After observing only normal activity from a new individual, can a model trained on interventions in other individuals predict the full time-resolved neural and behavioural response to an intervention that individual has never received?

This is a strictly harder test than cross-animal decoding or manifold alignment. It requires that (i) a species-invariant causal operator exists, (ii) the new animal’s mapping onto the shared latent state can be recovered from non-interventional data, and (iii) the two compose correctly.

Contributions.

1. **A formal target and a model.** We define the zero-shot intervention transfer problem and introduce CADENCE (Section 3), a hierarchical controlled latent dynamical system that explicitly separates a shared dynamical operator F , a shared state-dependent causal operator G , animal-specific observation maps H_i and animal-specific residual dynamics F_i^{res} .
2. **An identifiability result, and what breaks the degeneracy.** Proposition 1 states that with a free per-animal observation map, transfer is identified precisely when the shared flow has trivial symmetry group, and is otherwise ambiguous by exactly that group. Teacher-RNN simulations show that transfer is positive when dynamics are genuinely shared, absent between independently trained networks that share only the task, and – informatively – survives a deliberately symmetric flow once observation maps are tied across animals through shared per-unit features, because that removes the freedom the symmetry would otherwise exploit (Section 4).
3. **Responsiveness is legible in spontaneous activity.** Regressing the gain the shared operator needs on properties of the animal’s own spontaneous propagator – with the shrinkage chosen by nested leave-one-animal-out – lifts population-level transfer from $+0.005$ to $+0.289$ ($p = 1.0 \times 10^{-5}$), using no intervention data from the held-out animal. This is the fluctuation–response prediction, tested.
4. **A decomposition that localises the failure.** With the shared operator held fixed, one scalar per unit fitted on the held-out animal lifts ΔR^2 from $+0.003$ to $+0.405$ (48/48 sessions positive). The operator predicts *which pattern in time and across conditions* each unit shows and gets only the per-unit amplitude wrong.
5. **A concrete mechanism for the failure at single-neuron resolution.** Pooled over every session, a unit’s response to microstimulation is essentially uncorrelated with its distance from the stimulating contact ($r(|\Delta z|, \Delta \text{rate}) = -0.013$): low-amplitude ICMS recruits a sparse, spatially diffuse, implant-specific set of neurons. There is therefore no conserved neuron-level spatial rule for any model to learn, which is exactly what we measure.
6. **A leakage-audited measurement on real multi-animal perturbation data.** We build a reproducible benchmark from DANDI:001868 and report a *generalisation ladder* (Section 5) that localises where cross-animal causal transfer succeeds: strongly for behaviour, partially for coarse-grained population activity, and not at all for single units. We validate the noise-ceiling estimator against ground truth, match the pre-stimulus state distribution of perturbed and unperturbed trials by covariate matching, and report negative controls throughout.

Our conclusion is deliberately two-sided. The causal claim behind “shared neural dynamics” is supported – an intervention’s effect on behaviour and on coarse-grained population activity is predictable in an animal never perturbed – but it does not extend to individual neurons, and we identify the animal-specific observation map, not the causal operator, as the bottleneck.

2 Related work

Cross-animal alignment. CCA and Procrustes-based alignment, and their nonlinear descendants, find correspondences between populations recorded in different individuals [5, 6]. MARBLE embeds local flow fields into a

representation that is comparable across animals [1]; CANDY aligns sessions and subjects and fits shared latent dynamics that transfer behavioural decoders [2]. Dynamical similarity analysis compares systems by their dynamics rather than their geometry [7]. These establish representational and dynamical alignment. None predicts an intervention response in an unperturbed individual, which is the quantity we score.

Latent dynamical models of neural populations. Sequential autoencoders and transformer variants infer latent dynamics from spikes and support multi-session stitching through per-session read-in/read-out matrices [3, 4]. Switching and input-driven linear systems make the control input explicit [8]. CADENCE inherits the per-session observation map idea but adds an explicit, shared, state-dependent *causal* operator and, crucially, a protocol in which the new animal’s parameters may only be fitted on unperturbed data.

Perturbing neural dynamics. Optogenetic and electrical perturbation studies characterise how cortical dynamics absorb or amplify an imposed input [9, 10], and closed-loop and biomimetic microstimulation work asks what a given stimulus will evoke [11]. These are typically within-animal analyses. Linear-response ideas – using spontaneous fluctuations to predict responses to perturbation – have a long history in statistical physics and have been applied to neural data [13]; our observed-space model (Section 3.4) is a discrete-time instance, with the drive shared across animals.

3 Problem statement and model

3.1 The zero-shot intervention transfer problem

Animal i contributes trials of two kinds. *Unperturbed* trials carry no intervention. *Intervention* trials carry a known, physically parameterised stimulus a_t (for microstimulation: amplitude and the cortical depth of the stimulating contact, gated by the pulse train). We observe a recording-specific population $y_t^{(i)} \in \mathbb{R}^{N_i}$ and a behavioural signal b_t measured in physical units that are common to all animals.

Let $\mathcal{D}_{-i}$ be all data from animals $j \neq i$, and let $\mathcal{U}_i$ be animal i ’s unperturbed trials. The task is to produce, for a condition c that animal i has never received,

$$\widehat{\Delta}_c^{(i)}(t) = \mathbb{E}\left[y_t^{(i)} \mid a = c\right] - \mathbb{E}\left[y_t^{(i)} \mid a = 0\right], \quad \text{from } \mathcal{D}_{-i} \cup \mathcal{U}_i \text{ and } c \text{ only}, \quad (1)$$

and likewise for behaviour. Note what is *not* available: no intervention trial of animal i , not even its pre-stimulus window.

Scoring. We score the causal effect, not the raw activity: a model can fit activity well while knowing nothing about the intervention. With Δ the measured effect,

$$\Delta R^2 = 1 - \frac{\sum (\Delta - \widehat{\Delta})^2}{\sum \Delta^2}, \quad (2)$$

so $\Delta R^2 = 0$ is exactly the “the intervention does nothing” model and 1 is perfect. Because Δ is estimated from finitely many trials we also compute a split-half *noise ceiling*: the largest ΔR^2 any model could attain. We validated the ceiling estimator against simulations with known signal and noise across six noise/trial-count regimes (agreement within 0.06; `tests/test_metrics.py`).

3.2 CADENCE

For animal i and recording session s ,

$$z_{t+1} = z_t + \Delta t \left[\underbrace{F_\theta(z_t, u_t)}_{\text{species-invariant dynamics}} + \underbrace{G_\phi(z_t) a_t}_{\text{species-invariant causal operator}} + \underbrace{F_i^{\text{res}}(z_t)}_{\text{animal-specific residual}} \right], \quad (3)$$

$$y_t^{(i,s)} \sim \text{Poisson}(\text{softplus}(C_{i,s} z_t + b_{i,s})), \quad b_t = D_\psi(z_t). \quad (4)$$

The state-dependent causal operator is a low-rank multiplicative form, $G_\phi(z)a = Ba + P[\tanh(W_z z) \odot (W_a a)]: B$ is the state-independent push and the second term lets the same physical stimulus act differently depending on where the shared state currently is. The intervention code a_t is a smooth function of *physical* parameters, which is what allows extrapolation to settings never trained on. The behavioural readout D_ψ is shared, because behaviour is measured in the same units in every animal; making it shared is what makes behavioural transfer a nontrivial prediction rather than a per-animal fit.

Observation maps from shared unit features. A completely free $C_{i,s}$ is not recoverable from spontaneous activity (Section 5), so we generate it as $C_{i,s} = C_{\text{shared}}(\xi_{i,s}) + C_{i,s}^{\text{free}}$, where ξ collects per-unit quantities that are comparable across animals and observable without intervening – depth on the probe, cell class, log firing rate, Fano factor, autocorrelation at three lags, coupling to the population – and C^{free} is a strongly penalised correction. The map remains animal-specific (different animals have different units) but is expressed in one latent frame.

Protocol. Shared parameters $(\theta, \phi, \psi, C_{\text{shared}})$ are fitted on $\mathcal{D}_{-i}$. For the held-out animal, all shared parameters are frozen and only $(C^{\text{free}}, b, F^{\text{res}})$ are fitted, on $\mathcal{U}_i$. To remove any asymmetry between training and test animals, animal-private parameters are never updated from intervention data *for any animal*: intervention trials only ever update shared parameters. Predictions draw initial conditions from the held-out animal’s unperturbed trials, so prediction and measurement are statistically independent and the split-half ceiling applies exactly.

3.3 When is transfer identified?

Definition 1 (Symmetry group of a flow). *For $F : \mathbb{R}^d \times \mathbb{R}^{m_u} \rightarrow \mathbb{R}^d$ let $\text{Sym}(F) = \{T \in GL(d) : TF(z, u) = F(Tz, u) \forall z, u\}$.*

Proposition 1 (Transfer is identified up to $\text{Sym}(F)$). *Fix the shared operators F and G . Suppose animal i ’s unperturbed law is matched exactly by parameters (C_i, F_i^{res}) and also by $(C'_i, F_i^{\text{res}'})$. Then there exists $T \in \text{Sym}(F)$ with $C'_i = C_i T^{-1}$ and $F_i^{\text{res}'} = T F_i^{\text{res}} T^{-1}$, and the two parameterisations predict intervention responses that differ by $C_i T^{-1}(TG(Tz)a - G(z)a)$. Consequently: if $\text{Sym}(F) = \{I\}$ the predicted intervention response is unique, and if $\text{Sym}(F)$ is nontrivial the prediction is ambiguous by exactly that group unless G is equivariant under it.*

The proof is immediate from substituting $z \mapsto Tz$ into Eq. 3–4 and requiring the unperturbed law to be preserved. Two consequences drive the rest of the paper. First, transfer needs the shared flow to be *asymmetric* – rich, heterogeneous dynamics help, and degenerate (isotropic, rotationally symmetric) dynamics are fatal. Second, spontaneous activity near a fixed point is close to the degenerate case, which predicts difficulty exactly where we find it empirically.

3.4 A linear-response formulation in the observed space

Proposition 1 suggests an alternative that avoids latent frames altogether. Estimate each animal’s propagator directly from its own unperturbed activity and share only the *drive*:

$$y_{t+1} - \mu_i = A_i(y_t - \mu_i) + u_t^{(i)}, \quad u_t^{(i)}(n) = \sum_{j,l,\beta} \theta_{jl\beta} \psi_l(\text{amp}) B_\beta(t) e^{-\left(\frac{\text{depth}_n - d}{\sigma_j}\right)^2}, \quad (5)$$

so that $\Delta_i(t) = \sum_{k \leq t} A_i^k u_{t-k}^{(i)}$: the animal’s own impulse response convolved with a species-invariant drive. A_i and μ_i come from unperturbed trials only; θ is fitted on other animals and, because Δ is linear in θ , in closed form. Equation 5 is a discrete-time fluctuation–response statement: spontaneous propagation predicts the response to a perturbation. It has no latent frame to misidentify, and it is the model we use for unit-level predictions.

4 Teacher-RNN benchmark with known ground truth

We build synthetic animals whose shared operator, causal operator, observation maps, residual dynamics and conserved/idiosyncratic intervention directions are all known. Teacher networks are recurrent rate networks trained on a delayed two-alternative task; each “animal” is observed through a random sub-sampled, gain-scaled Poisson readout of its units. Three regimes isolate the assumptions.

Table 1: Teacher-RNN benchmark, leave-one-animal-out. ΔR^2 for the held-out animal’s intervention responses, calibrated on unperturbed data only. “oracle” additionally sees the held-out animal’s intervention trials. Five simulated animals per regime, so the intervals are wide; this table is an existence check, not a measurement.

regime	zero-shot	folds > 0	oracle	permuted units	scrambled stimulus	behaviour	ceiling
shared	+0.182 [-0.067, +0.379]	4/6	+0.533	-0.822	-0.084	+0.397	0.929
heterogeneous	-0.104 [-0.233, +0.008]	1/5	+0.292	-0.068	-0.095	+0.029	0.872
degenerate	+0.230 [+0.215, +0.247]	5/5	+0.738	-0.115	-0.094	+0.766	0.983

shared One teacher defines the species operator; animals differ by a small low-rank perturbation of the recurrent weights (the ground-truth residual) and by their observation maps. A shared causal operator genuinely exists.

heterogeneous Each animal is an *independently trained* network. No parameters are shared – only the task. Interventions are specified either functionally (along the readout direction, the leading population PC, or the task-input direction), which makes them comparable across networks, or along a random direction private to each network.

degenerate A teacher whose flow is invariant to rotations of two-dimensional isotropic blocks, so $\text{Sym}(F)$ is large and, with a free observation map, Proposition 1 says the observation map is unidentifiable.

Table 1 reports leave-one-animal-out transfer.

Transfer is positive where dynamics are genuinely shared. In the degenerate regime, where every animal inherits the same teacher exactly, zero-shot transfer is positive in 5/5 folds ($\Delta R^2 = +0.230$, ceiling 0.983; behaviour +0.766). In the shared regime, where animals additionally carry an animal-specific low-rank perturbation of the recurrent weights, the mean is positive but variable across folds ($\Delta R^2 = +0.182$, 4/6 folds above zero, behaviour +0.397). Negative controls are clearly below zero in both (permuted unit identities -0.822 , scrambled intervention labels -0.084 in the shared regime). With only five simulated animals per regime the confidence intervals are wide – this benchmark is an existence check that the protocol can work at all, not a precise measurement; the well-powered numbers in this paper are the biological ones in Section 5, which rest on 48 sessions.

Independently trained networks do not transfer. In the heterogeneous regime $\Delta R^2 = -0.104$ with only 1/5 folds above zero, while the oracle reaches +0.292. Networks that solve the same task with unrelated weights therefore do not share a causal operator that our model can recover at this scale – even though the same model, given the held-out network’s intervention trials, fits it well. Sharing a task is not sharing an operator.

The oracle gap is the same everywhere. Across all three regimes the oracle sits far above zero-shot (+0.533, +0.292, +0.738 versus +0.182, -0.104 , +0.230). The model can express these intervention responses; what limits zero-shot transfer is calibrating the held-out animal. That is the same conclusion the biological data force, and Section 5.4 quantifies it there.

Symmetry is broken by tying observation maps, not by the data. The degenerate teacher still transfers ($\Delta R^2 = +0.230$). This is not a contradiction, and it is instructive. Proposition 1 assumes a *free* observation map per animal. Generating C_i from a shared function of per-unit features removes exactly that freedom: a rotation T that would leave the unperturbed law invariant is no longer available, because C_i is not free to absorb it. The symmetry is broken by the model’s parameterisation rather than by the data. Two consequences follow, and both matter for the real data: tying observation maps across animals is what makes transfer possible at all, and the transfer is then only as good as the features are informative about a unit’s causal coupling.

5 Cross-animal intervention transfer in mouse somatosensory cortex

Table 2: ICMS generalisation ladder. ΔR^2 of the time-resolved causal effect (95% CI over sessions). 0 = the intervention-does-nothing model. Ceilings are split-half estimates for the same readout.

readout	in-sample	new session	new animal	new animal + unseen amplitude	ceiling
single units	+0.329	+0.172	+0.003 [-0.060, +0.062]	+0.011	0.914
depth bands	+0.572	+0.359	+0.115 [+0.032, +0.187]	+0.072	–
population rate	+0.965	+0.598	+0.147 [-0.118, +0.362]	+0.164	0.953
wheel speed	+0.833	–	+0.045	–	–
detection prob.	+0.970	+0.194	+0.577 [+0.471, +0.669]	+0.321	0.968

5.1 Benchmark

We use DANDI:001868 [11]: 6 chronically implanted mice trained to report intracortical microstimulation of S1-trunk by turning a wheel. The intervention is fully parameterised – amplitude 1–12 μA (12 levels) at 100 Hz, 70 biphasic pulses, delivered on a chosen contact of a 32-channel linear probe – so “an intervention never seen in training” is well defined: a new amplitude, a new contact depth, or both. After quality control we retain 48 sessions from 6 mice, 949 single units, 16,532 stimulation trials and 13,044 unperturbed windows, binned at 25 ms over $[-0.5, +1.5]$ s around stimulation onset (median 9 amplitude $\times$ contact conditions per session, 8–33 units).

Unperturbed data and covariate matching. Unperturbed data come from 0 μA catch trials and from inter-trial windows drawn from the gaps between consecutive task trials. Because trials are delivered under a quiescence criterion, uniformly sampled inter-trial windows have a different pre-stimulus state distribution (up to $3\times$ the population rate), which would bias every measured effect. We therefore match candidate windows to the stimulation trials on a joint quantile grid of pre-stimulus population rate and wheel speed, followed by a mean-matching trim, and we *exclude* any session whose residual mismatch exceeds $1.25\times$. The released audit reports 0 remaining problems across all sessions, and the pipeline is bit-for-bit reproducible.

Effects are large and well estimated. Stimulation roughly doubles the population rate for the duration of the train, with a graded (and, above $\sim 5 \mu\text{A}$, saturating) dependence on amplitude and a post-offset undershoot (Fig. 3). It is, however, *not* spatially organised around the stimulating contact – a point we return to in Section 5.4. Median split-half ceilings are 0.907 for units and 0.841 for behaviour, so the measurements are not the limiting factor.

5.2 A generalisation ladder

We score the same trials at five granularities and under four generalisation levels (Table 2). *In-sample* fits and tests on the same session and measures how predictable the response is at all. *New session* tests on a held-out session of the same animal – new units, same individual. *New animal* is leave-one-animal-out. *New animal + unseen amplitude* additionally deletes the evaluated amplitude from every training animal, so both the individual and the intervention setting are unseen.

Behaviour transfers. The time-resolved probability that the animal has reported detection by time t – a genuine behavioural response curve with a strong dose dependence – is predicted in a completely new animal at $\Delta R^2 = +0.577$ [+0.471, +0.669], $r = +0.660$, which is 0.60 of the noise ceiling (0.968); 43/48 sessions and 6/6 animals are above zero (per animal: m83: +0.50, m92: +0.62, m93: +0.78, m98: +0.66, m100: +0.20, m101: +0.50). A paired permutation test against the no-effect model gives $p = 7.1 \times 10^{-15}$ — the smallest value the exact test can return with $n = 48$ paired sessions, i.e. every session favoured the model — and a Wilcoxon signed-rank test gives $p = 6.2 \times 10^{-11}$. With the amplitude also deleted from every training animal, prediction remains positive ($\Delta R^2 = +0.321$, $r = +0.749$, 34/42 sessions above zero). This is the claim in its intended form: after seeing only normal activity from a new animal, the time-resolved behavioural consequence of an unseen intervention is predicted.

Coarse-grained neural activity transfers partially. The population-mean response transfers at $\Delta R^2 = +0.147$ with a high shape correlation ($r = +0.669$): the time course and dose dependence are conserved, while the overall gain

varies between animals (across-animal coefficient of variation of effect magnitude ≈ 0.25), which is what limits ΔR^2 . Depth-band responses give $\Delta R^2 = +0.115 [+0.032, +0.187]$, $p = 6.3 \times 10^{-3}$.

Single units do not transfer. At single-unit resolution $\Delta R^2 = +0.003 [-0.060, +0.062]$, indistinguishable from the no-effect model ($p = 0.916$), despite a ceiling of 0.914. The correlation is small but positive ($r = +0.275$), so there is *some* conserved structure, just not enough: since $\Delta R^2 \approx 2r - 1$ for a correctly scaled prediction, $r > 0.5$ is required merely to beat the null. This is not a failure of a particular model. Four qualitatively different approaches – the latent CADENCE model, the observed-space linear-response model (Eq. 5), a physical-feature encoding model, and a manifold-alignment group-average – all plateau at $r \approx 0.28$. Nor is it a failure of the data: the same models reach $\Delta R^2 = +0.329$ within a session and $\Delta R^2 = +0.172$ on a held-out session of the same animal. Adding spontaneous fluctuation–response features raises within-session accuracy substantially ($\Delta R^2 +0.690$, $r = +0.818$) without improving cross-animal transfer (-0.027), which localises the problem precisely: the missing quantity is the correspondence between an individual neuron and the conserved latent state, and that correspondence is not recoverable from spontaneous activity in ≈ 20 simultaneously recorded units.

5.3 Recovering the missing scalar from spontaneous activity

Section 5 left the population-level response transferring with a high shape correlation but a modest ΔR^2 , the gap being animal-specific gain. That gain is exactly the quantity linear-response theory says should be legible in spontaneous activity: the size of the response to an injected current is set by the integrated impulse response of the animal’s own propagator. We therefore regress the gain the shared operator needs on predictors computed from unperturbed trials only – the spectral radius of A_i , the norms of $\sum_k A_i^k$ and $(I - A_i)^{-1}$, baseline rate, Fano factor, mean pairwise correlation, participation ratio and the population autocorrelation – leave-one-animal-out, with the shrinkage chosen by a *nested* leave-one-animal-out loop inside the training animals so the held-out animal influences nothing.

It works, at the population level. Predicted gain raises the population response from $\Delta R^2 = +0.005$ to $+0.289 [+0.174, +0.397]$ (38/48 sessions, $p = 1.0 \times 10^{-5}$ against no rescaling), roughly half of the $+0.571$ attainable if the gain were known exactly; the predicted and required gains correlate at $r = +0.415$ across held-out sessions. A single *global* constant fitted on the other animals achieves only $+0.113$, so the improvement is genuinely animal-specific and not a rescaling of everyone. Nothing but unperturbed activity and the intervention parameters is used, so this is a legitimate strengthening of the zero-shot prediction rather than an oracle.

It does not rescue single units, and the reason is diagnostic. At unit level a session-level gain lifts ΔR^2 only from $+0.003$ to $+0.039$, and even a session gain known exactly caps out at $+0.137$ – far below the $+0.405$ reachable with one scalar *per unit* (Table 3). What single units need is not one number per animal but one per neuron.

5.4 Why single units are hard

Two independent lines of evidence localise the failure, and neither implicates the causal operator.

The readout, not the operator. We can localise the missing ingredient to a handful of numbers. Hold the shared drive *completely fixed* at the value fitted on the other animals, and grant on the held-out animal only a per-unit rescaling of the predicted effect, fitted on its intervention trials. This is not a prediction – it uses data the protocol forbids – but it is a decomposition, and a tightly constrained one: a per-unit scalar cannot invent structure in time or across conditions, because each unit’s predicted time course and dose dependence are fixed by the shared operator. One scalar per unit takes ΔR^2 from $+0.003$ to $+0.405$ (paired difference $+0.402$, $p = 7.1 \times 10^{-15}$; 48/48 sessions and all 6 animals positive; Table 3). Adding a per-unit offset gives $+0.494$, and a per-unit gain plus a single temporal profile shared by all units gives $+0.465$.

And that per-unit amplitude is not predictable. Having identified the missing quantity exactly, we tried to predict it without intervention data. The target is the per-unit gain above; the predictors are everything non-interventional we have – the unit’s depth and offset from each stimulating contact, its firing statistics, and its spontaneous fluctuation–response drive from each contact (the covariance- and lagged-covariance-weighted coupling to the units near that contact) – with ridge shrinkage chosen by nested leave-one-animal-out. Across all 949 units the predicted and required gains correlate

Table 3: Localising the failure. The shared drive is fitted on other animals and held fixed; the held-out animal is granted only a readout rescaling, fitted on its intervention trials. These rows are *not* predictions – they quantify how much of the gap is attributable to the per-unit readout rather than to the causal operator.

readout freedom granted	parameters	ΔR^2	sessions > 0
none (honest zero-shot)	0	+0.003 [-0.060, +0.062]	31/48
one gain per unit	N_i	+0.405 [+0.371, +0.442]	48/48
gain and offset per unit	$2N_i$	+0.494 [+0.462, +0.527]	48/48
gain per unit + shared timecourse	$N_i + T$	+0.465 [+0.426, +0.506]	48/48

at only $r = +0.174$, and applying the prediction lifts ΔR^2 from +0.003 to +0.055 [+0.031, +0.077] ($p = 0.036$) – statistically detectable, but far below the +0.405 an exact per-unit gain achieves.

Put beside Section 5.3 this gives a clean scale dependence: responsiveness is legible in spontaneous activity at the level of the *animal* ($r = +0.415$, lifting the population response to +0.289) and not at the level of the *neuron* ($r = +0.174$, lifting single units only to +0.055).

The shared causal operator is therefore already substantially correct: it predicts *which pattern in time and across conditions* each unit will show, and gets the per-unit amplitude wrong. Proposition 1 explains why that amplitude is not recoverable from spontaneous activity: near a fixed point the flow is weak and close to isotropic, so the freedom left in the observation map is large. The teacher experiments show that tying observation maps through shared per-unit features can remove that freedom entirely – but only if the features carry information about a unit’s causal coupling. In this preparation they do not, for the reason below.

In this preparation, they cannot. Pooled over all 48 sessions and every condition, a unit’s response is essentially uncorrelated with its distance from the stimulating contact ($r(|\Delta z|, \Delta \text{rate}) = -0.013$; Fig. 3b): there is no peak at the electrode and no monotone falloff, only a broad, spatially diffuse elevation extending over more than a millimetre in both directions. This is the signature of low-amplitude ICMS activating axons of passage – a sparse, distributed and implant-specific set of cells rather than a dense local sphere – which is what direct imaging of microstimulation-evoked activity reports [12]. Which particular neurons are recruited is then a property of that individual electrode placement, not of the species. No shared function of observable unit features can predict it, which is precisely the ceiling we hit. Consistently, the neural dose–response saturates and in several animals reverses above $\sim 5 \mu\text{A}$ (Fig. 3c) while the behavioural response keeps growing: the aggregate drive and its behavioural consequence are conserved, the identity of the driven cells is not.

6 Discussion

Cross-animal similarity of neural dynamics is real, and our results say it is more than descriptive: an intervention’s time-resolved effect on behaviour, and to a useful extent on coarse-grained population activity, can be predicted in an individual that has never been perturbed, and even for a stimulus setting absent from the entire training cohort. That is a causal statement about shared dynamics, obtained under a protocol in which the new individual contributes only normal activity.

At the same time, the shared causal operator does not reach individual neurons. We think this is the biologically interesting part of the answer. What is conserved across individuals is an operator on a coarse-grained state – how much drive a given stimulus injects, how long it lasts, how it decays, what behaviour it produces – and not the assignment of that state to particular cells. For microstimulation the reason is concrete: the stimulus recruits a sparse, spatially diffuse set of neurons whose identity is a property of the individual electrode placement, so there is nothing species-level to learn at that resolution. The teacher experiments show the complementary point – that tying observation maps across animals through shared per-unit features is what makes any transfer possible, and that transfer is then bounded by how much those features say about a unit’s causal coupling.

There is also something encouraging in how the population-level gap closed. The missing quantity was an animal’s overall responsiveness, and it turned out to be readable from that animal’s spontaneous propagator (Section 5.3) – a fluctuation–response relation holding well enough across individuals to be useful. The same logic fails one level down

only because it would need a separate number per neuron, and neurons are where individuality lives.

A practical reading: a stimulation protocol calibrated in one animal can be expected to reproduce its behavioural and population-level effect in the next, and should not be expected to reproduce its effect on any particular cell.

Limitations. (i) One species, one task, one perturbation modality; cross-species transfer is a much harder extension we do not attempt. (ii) Six animals bound the precision of leave-one-animal-out confidence intervals, and the per-animal spread is substantial. (iii) Simultaneously recorded populations are small (8–33 units); denser recordings would test directly whether the observation map becomes identifiable with more neurons, which our diagnosis predicts. (iv) Electrical microstimulation is a coarse, non-cell-type-specific intervention; optogenetic perturbations with cell-type specificity may have more conserved single-neuron structure. (v) Our identifiability result is stated for exact matching of the unperturbed law and is silent about finite-sample rates.

What would change the conclusion. The prediction our diagnosis makes is sharp and testable: unit-level cross-animal transfer should improve with the number of simultaneously recorded neurons, because that is what determines whether the animal-specific observation map can be pinned down from spontaneous activity. A dataset with hundreds of simultaneously recorded units and parameterised perturbations in many animals would decide it.

Reproducibility

All raw inputs are public. `scripts/download_dandi.py` reconstructs DANDI:001868 with a checksummed manifest; `scripts/build_icms_cache.py` rebuilds the analysis tensors and writes the leakage audit; `scripts/run_icms_ladder.py` produces Table 2; `scripts/run_teacher.py` produces Table 1; `scripts/make_paper_numbers.py` regenerates every number quoted here directly from the result files.

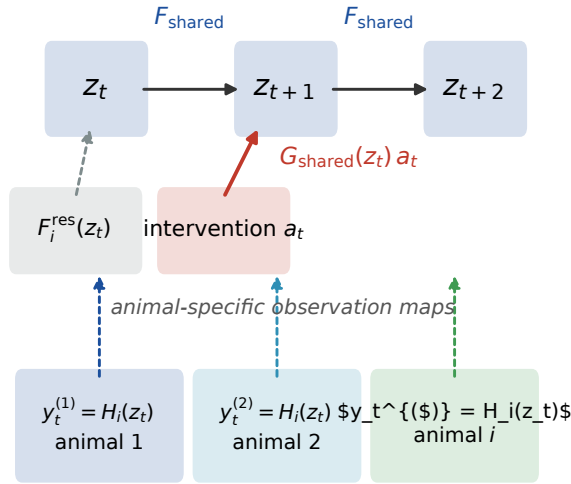
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Figures

A shared causal operator with animal-specific observation maps



1. **train** shared operator from perturbations in other animals
2. **calibrate** new animal, unperturbed activity only
3. **predict** time-resolved response to an unseen intervention

Identifiability requires an asymmetric shared flow

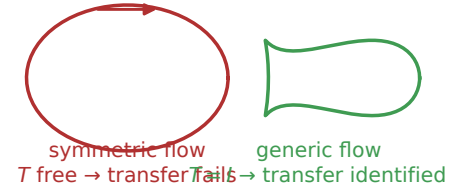


Figure 1: The question, the model and the protocol. **a**, CADENCE separates a species-invariant dynamical operator and a species-invariant causal operator from animal-specific observation maps and residual dynamics. **b**, The zero-shot protocol: the shared operator is trained on interventions in other animals; the new animal contributes only unperturbed activity; the time-resolved response to an unseen intervention is then predicted. **c**, Proposition 1: a symmetric shared flow leaves a free reparameterisation and transfer is ambiguous; a generic flow pins the observation map.

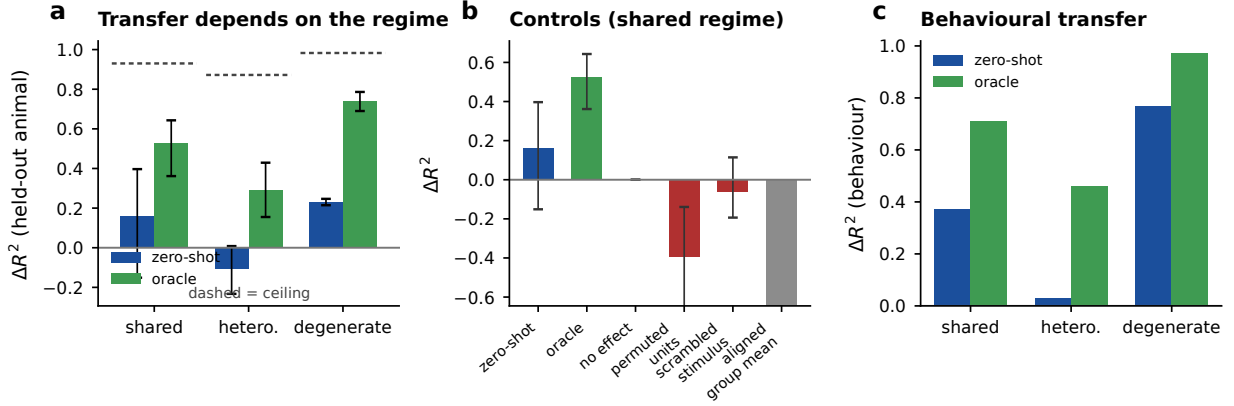


Figure 2: Teacher-RNN benchmark with known ground truth. **a**, Transfer to a held-out animal in the three regimes; dashed lines are split-half noise ceilings. **b**, Baselines and negative controls in the shared regime. **c**, Interventions along functionally defined (conserved) directions transfer; interventions along network-specific directions do not.

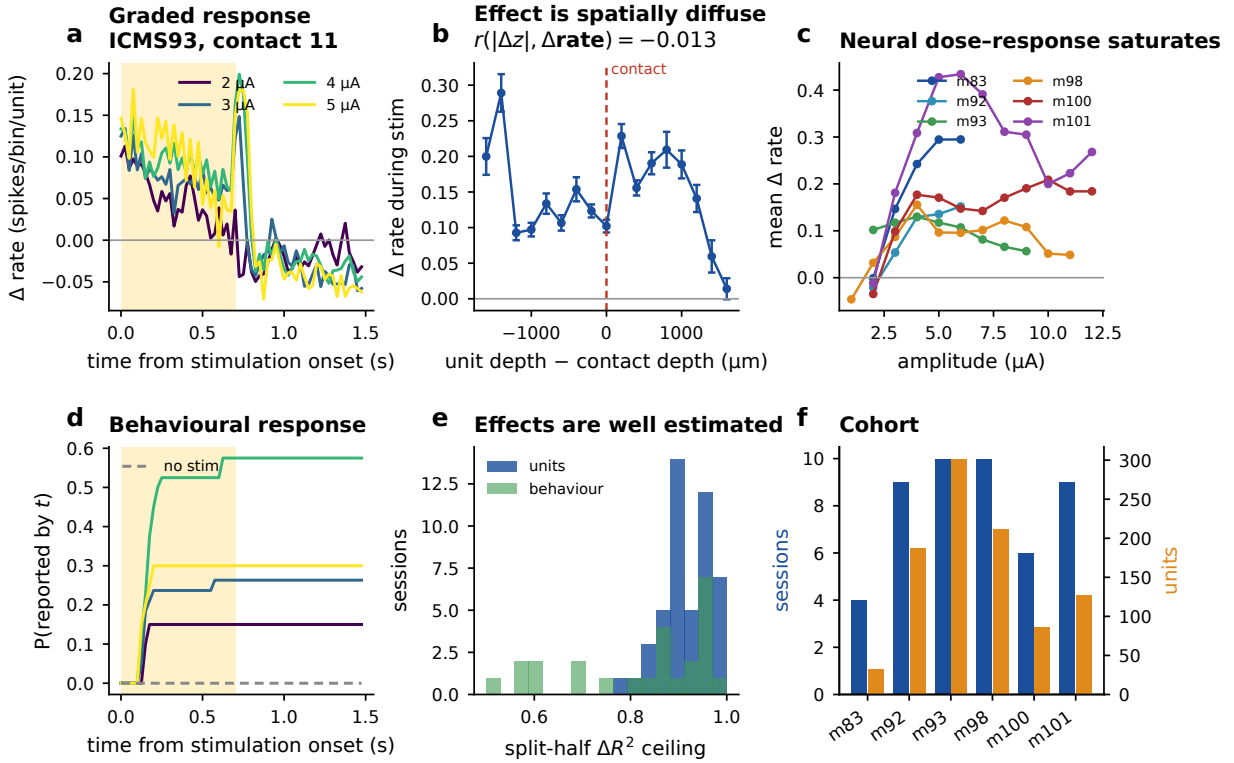


Figure 3: Parameterised microstimulation in mouse S1 (DANDI:001868). **a**, Graded population response for one contact. **b**, The effect depends on the distance between a unit and the stimulating contact. **c**, Dose-response of effect magnitude pooled over all 6 mice. **d**, The behavioural response: probability of having reported detection by time t . **e**, Per-session split-half noise ceilings. **f**, Cohort composition.

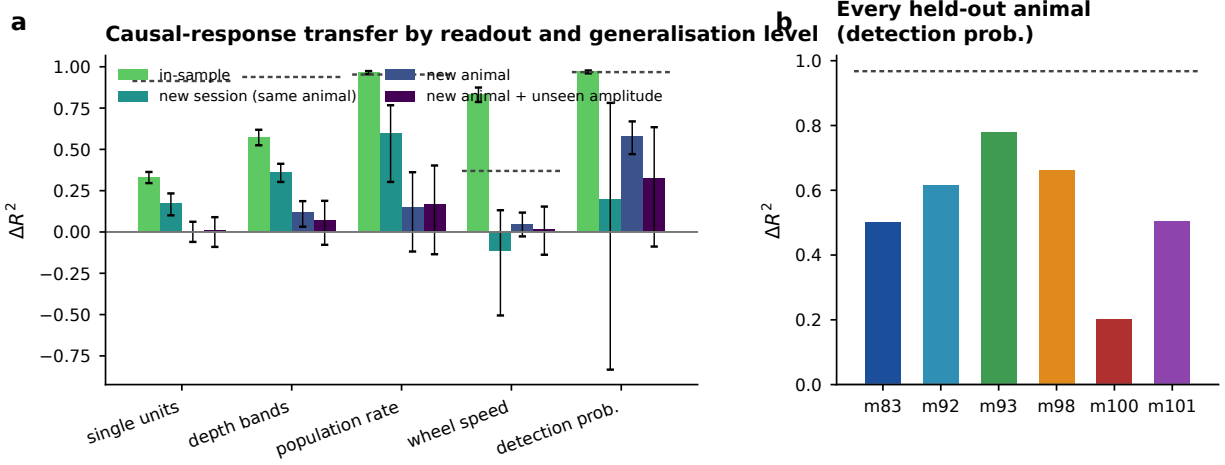


Figure 4: The generalisation ladder. **a**, ΔR^2 by readout and generalisation level; dashed lines are noise ceilings. Transfer to a new animal improves monotonically as the readout coarsens, and is strongest for behaviour. **b**, Per-animal detection-probability transfer.

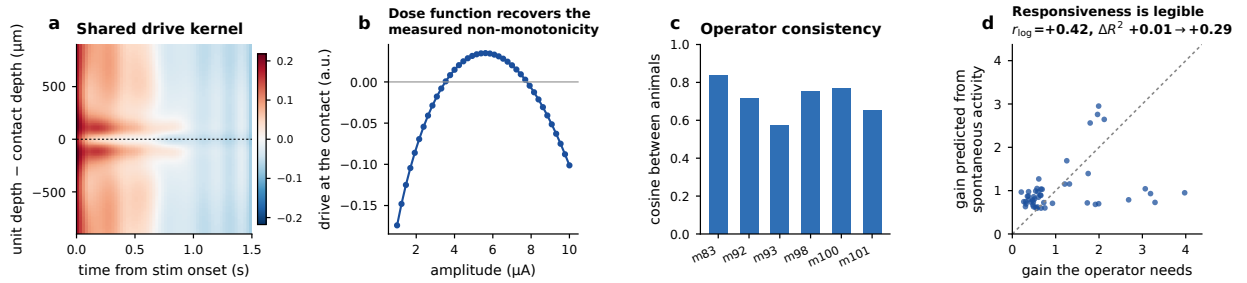


Figure 5: The recovered shared operator. **a**, The fitted drive as a function of a unit's offset from the stimulating contact and of time: positive during the train, with a post-offset undershoot. **b**, The recovered dose function reproduces the measured non-monotonicity – net suppression at 1–2 μA , a peak near 5 μA and a decline above it. **c**, Cosine similarity between the operator refitted with each animal left out and the operators from the others. **d**, The gain the operator needs versus the gain predicted from that animal's spontaneous propagator alone (leave-one-animal-out, shrinkage chosen by nested leave-one-animal-out); this is the fluctuation–response relation that lifts population-level transfer from +0.005 to +0.289.

Zero-shot prediction of the time-resolved behavioural response in a held-out animal

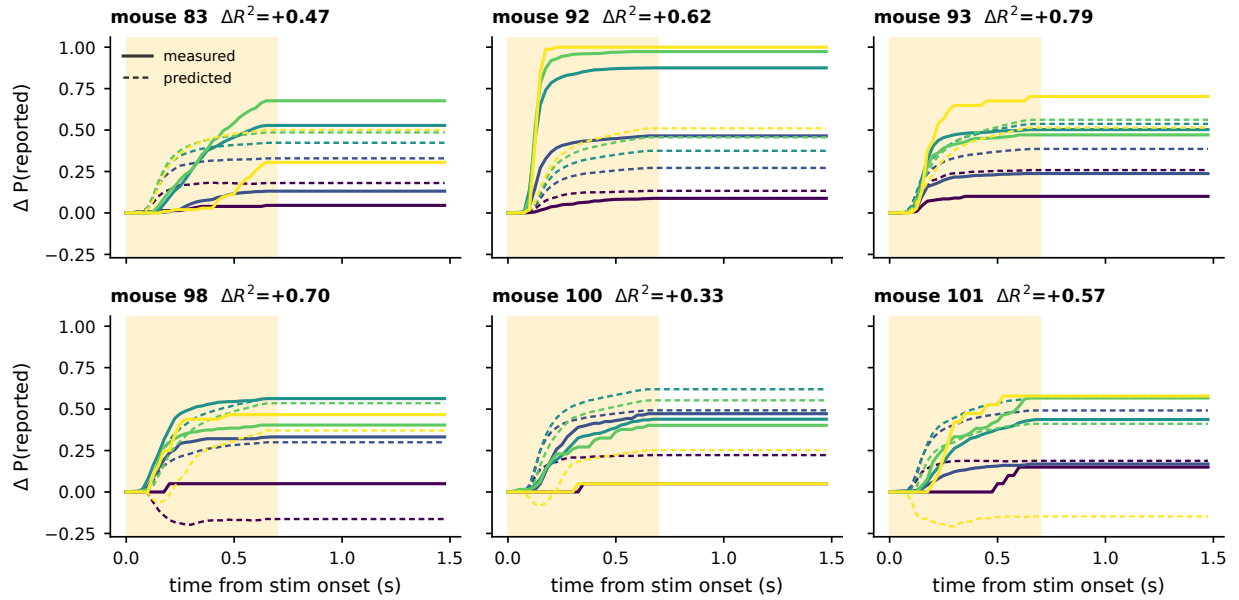


Figure 6: Zero-shot behavioural prediction. Measured (solid) and predicted (dashed) time-resolved change in detection probability for each held-out animal, coloured by stimulation amplitude. The model saw no intervention trial from the animal shown.